PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila











PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Beyond Amino Acids: Bioactive Peptides as Precision Modulators in animal nutrition

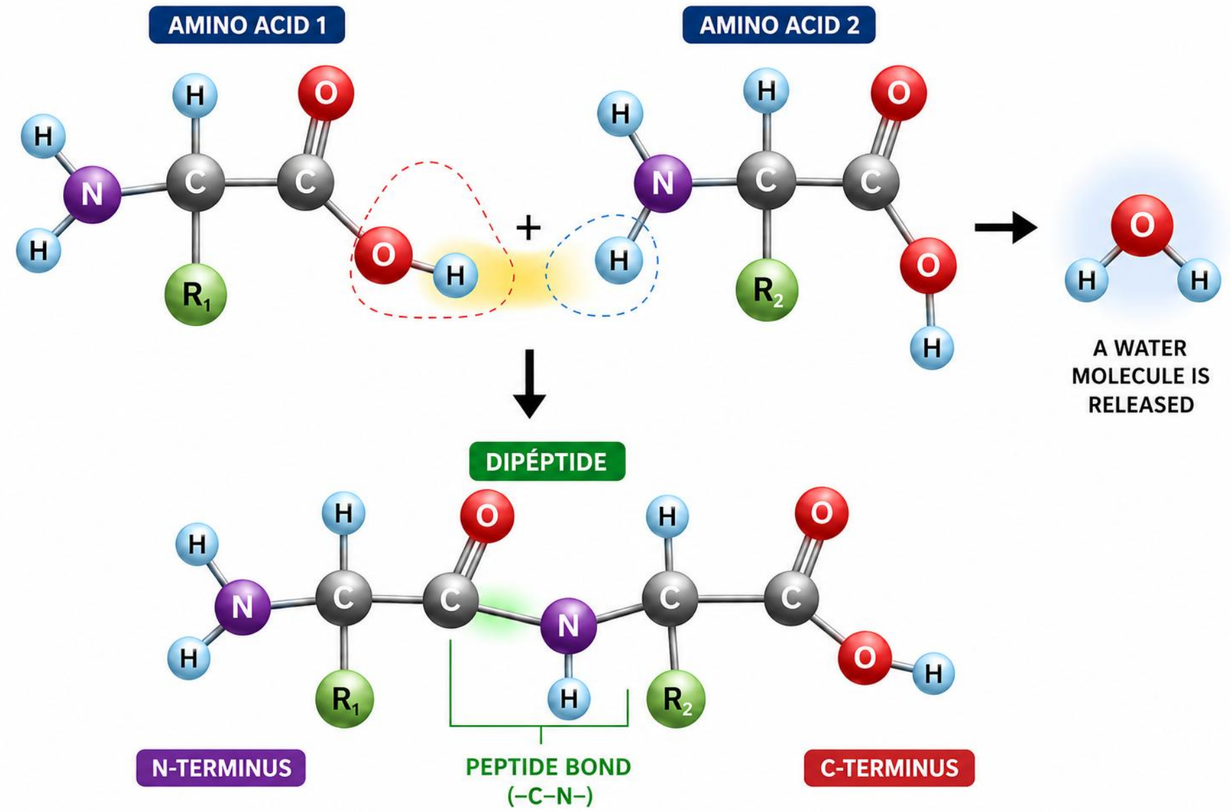
Xavier Córdoba
Samuel Correa
Marina Sarmiento

Basic structure of Proteins

- Proteins are **polymers of amino acids** linked by peptide bonds

- An amino acid is the basic protein molecule composed of a carbon attached to:

  - **Amino** group (NH_2)
  - **Carboxyl** Group (COOH)
  - **Hydrogen** Atom (H)
  - **Side Chain** (R)



Types of Amino Acids

There are **21 amino acids**
(including selenocysteine)

The **side chain (R) determines the characteristic properties** for each, and these properties contribute to structure and function of the protein.

1-letter	3-letter	Name	Functional Property
M	Met	Methionine	Hydrophobic; sulfur-containing
F	Phe	Phenylalanine	Aromatic, hydrophobic
W	Trp	Tryptophan	Aromatic, hydrophobic
Y	Tyr	Tyrosine	Aromatic; weakly polar
P	Pro	Proline	Rigid; constrains the peptide backbone
G	Gly	Glycine	High structural flexibility
C	Cys	Cysteine	Thiol; forms disulfide bonds
S	Ser	Serine	Polar, uncharged
T	Thr	Threonine	Polar, uncharged
N	Asn	Asparagine	Polar, uncharged
Q	Gln	Glutamine	Polar, uncharged
D	Asp	Aspartic acid	Negatively charged
E	Glu	Glutamic acid	Negatively charged
K	Lys	Lysine	Positively charged
R	Arg	Arginine	Positively charged
H	His	Histidine	Weakly basic; partially charged
U	Sec	Selenocysteine	Redox-active selenol group
A	Ala	Alanine	Hydrophobic
V	Val	Valine	Hydrophobic
L	Leu	Leucine	Hydrophobic
I	Ile	Isoleucine	Hydrophobic

Proteins do far more than supply amino acids

Antioxidant Defense: Critical and underappreciated role in antioxidant defense. e.g. GPx / Glutathione Peroxidase

Transport of Molecules: Transporting molecules across cellular membranes and throughout the body. e.g. hemoglobin

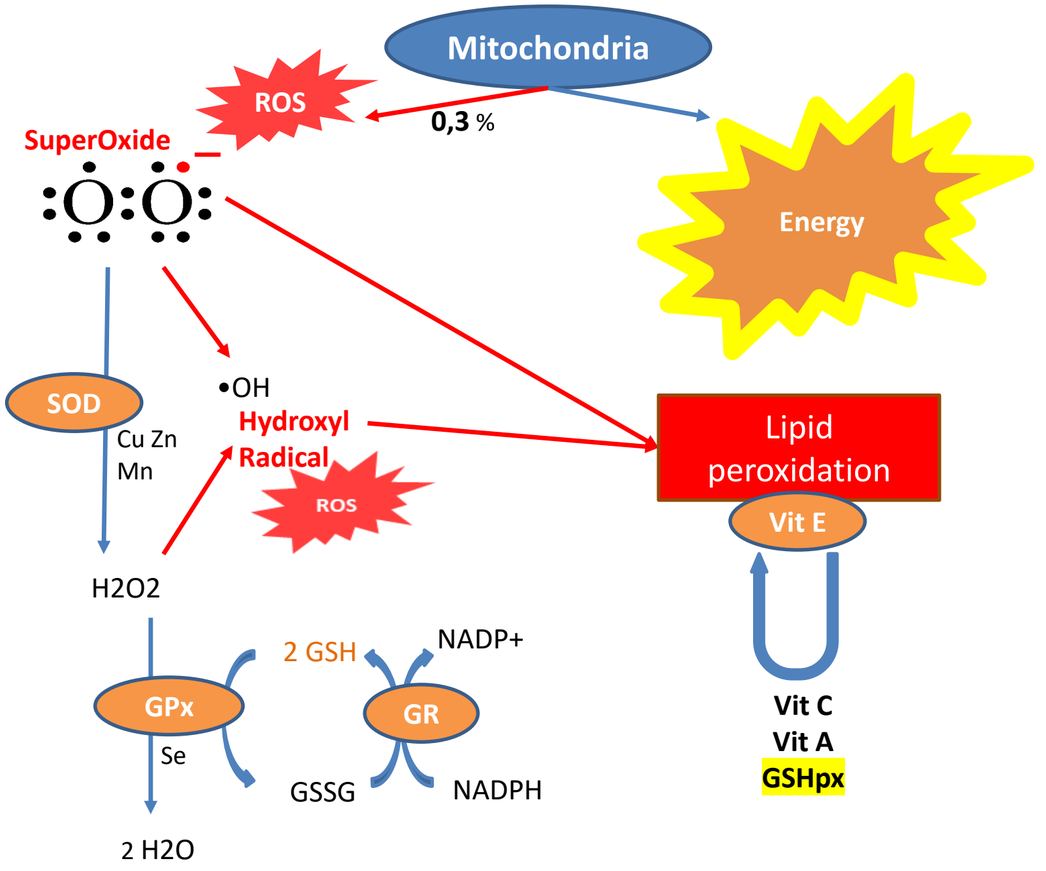
Enzymatic Activity: Proteins act as enzymes, catalyzing biochemical reactions that are essential for life. e.g. pepsin

Hormonal Regulation: Proteins also function as hormones that regulate various physiological processes. e.g. Insulin

Immune Response: Proteins defend the body against pathogens. e.g. Defensins (antimicrobial proteins)

Storage: Stores essential substances. E.g. ovalbumin

Flavor Enhancement: Proteins contribute to flavor through the release of amino acids like glutamate and aspartate



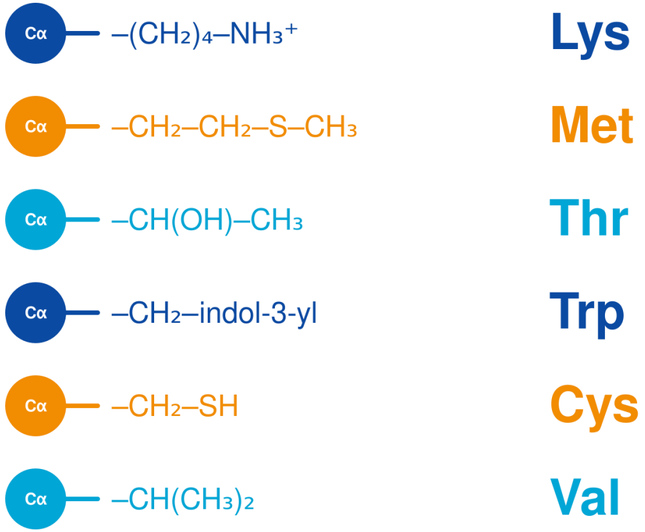
SOD:	Superoxide Dismutase
GPx:	Glutathione Peroxidase
GR:	Glutathione Reductase
GSH:	Glutathione

... are Proteins !

Biological value of a protein

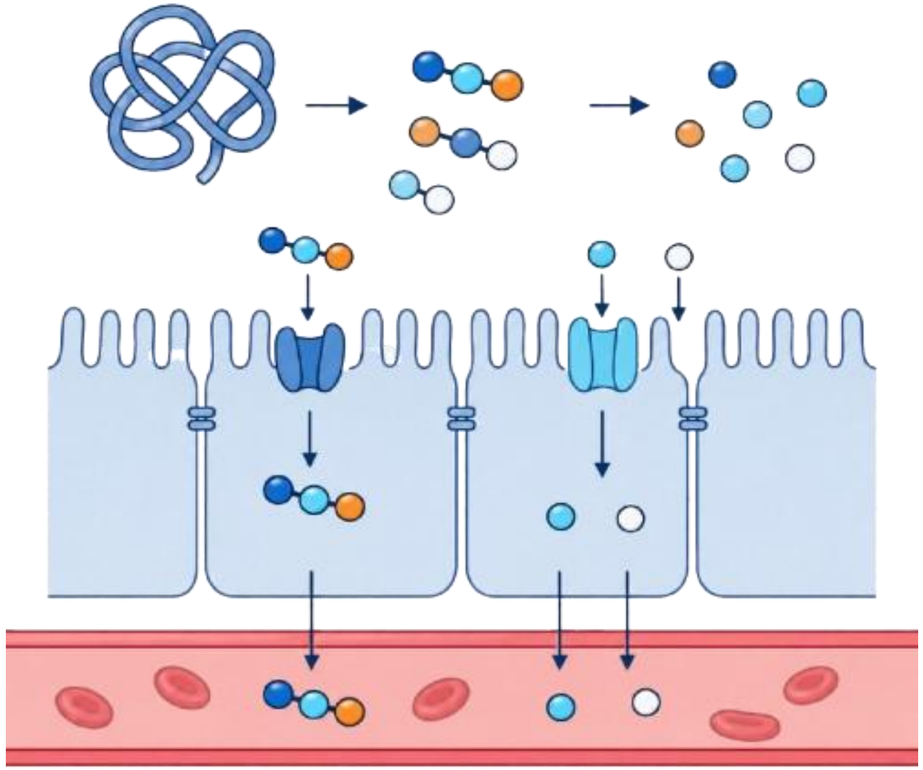
Amino acid profile

What does it provide?



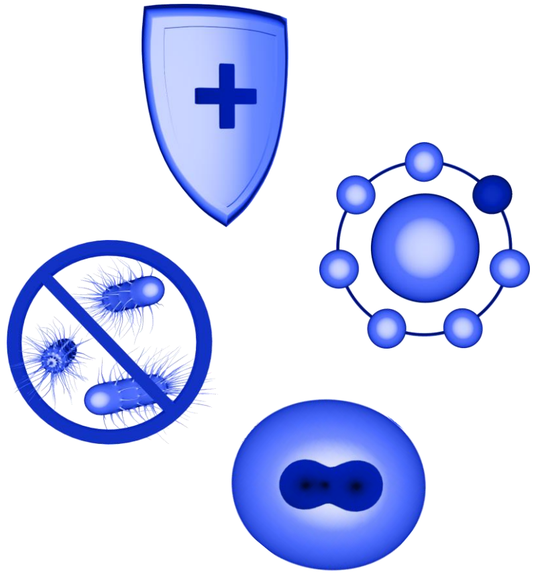
Digestibility

What becomes available?



Functionality

What can it modulate?



What is a Peptide

Proteins

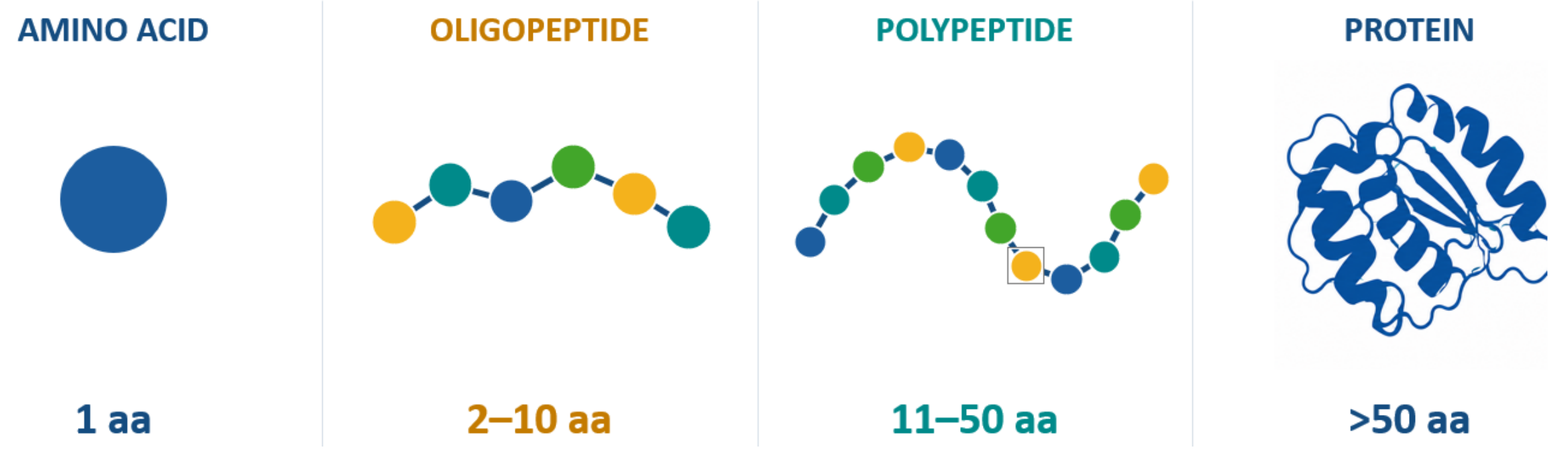
Functionality is more diverse and complex in proteins than in peptides.

Proteins function is **very sensitive** to thermal processes, pH variations and hydrolysis.

Peptides

They tend to have more **limited functionalities.**

Peptides are **often more resistant** than intact proteins to **heat and pH changes**, and in function of the origin could be more resistant to peptidases.



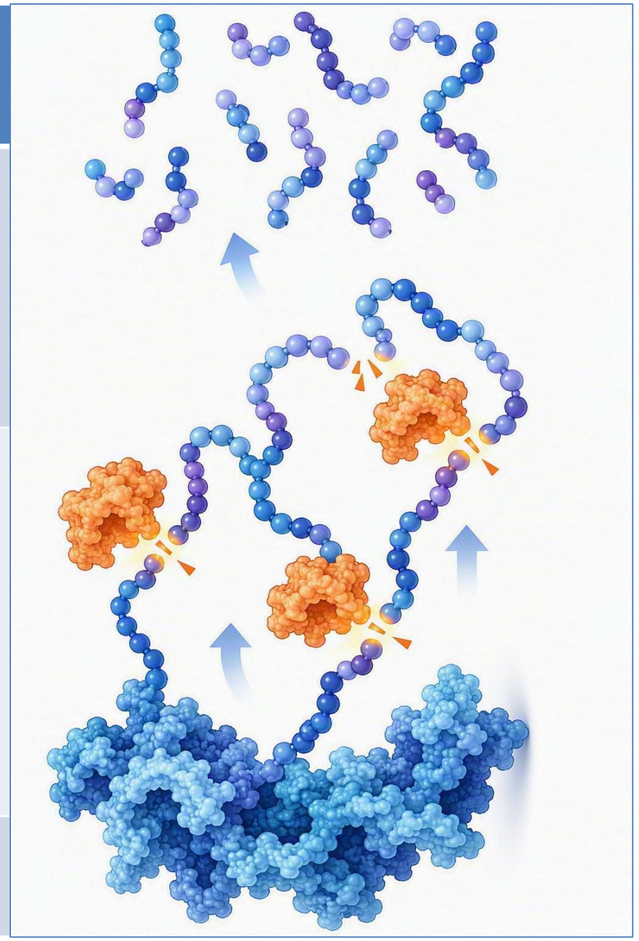
5-20 aa
Typical Molecular Range of Bioactive Peptides

Molecular Weight
550 – 2200 Da

aa: amino acids

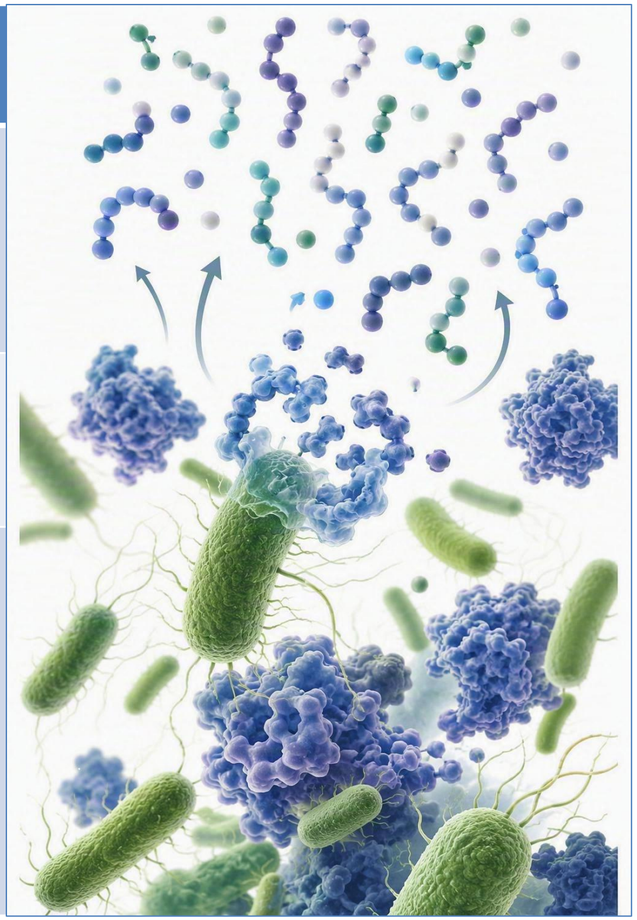
Methods to produce Bioactive Peptides

Enzymatic Hydrolysis	
Select a tissue with high potential	→ Certain **Animal origins are good candidates.** → Some **glandular and collagen-rich** tissues are valuable sources with high bioactive potential.
Controlled process of hydrolysis	→ Type of **enzyme, time, pH, temperature** are essential → **Insufficient hydrolysis** may fail to reach the target MW while **over hydrolysis** can degrade bioactive peptides.
System of ultrafiltration	→ **Combined** with a correct time of hydrolysis it's a good strategy to increase the success.



Methods to produce Bioactive Peptides

Fermentation (GMO Free)	
Select a Natural Peptide-Producing Strain	→ Certain microorganisms naturally produce bioactive peptides, **particularly antimicrobial** bacteriocins.
Controlled Fermentation	→ Fermentation conditions—strain, medium, pH, temperature, and time—are optimized to maximize peptide biosynthesis and secretion.
Recovery and Purification	→ Cells and insoluble material are removed, and the peptides are concentrated and purified **using membrane filtration or chromatography.** → **If used, care must be taken with the excess of hydrolysis**



Recombinant Production of Specific Peptides (GMO)

The gene encoding the target peptide can be introduced or optimized in a suitable production microorganism using recombinant DNA or CRISPR-based technologies. The production strain and final product **require case-by-case safety assessment**.

Bioactive Peptides: Basic Characteristics

Bioactive function	Typical Length (aa)	Typical Window (Da)	Possible Range (Da)	Some key features
Antimicrobial (AMP)	12–20	**1500–3000**	1000–4000	Amphipathic; charge +; 30–60% hydrophobic
Antioxidant	3–8	**400–1000**	200–1500	Aromatics (Tyr, Trp), His, Cys, Met; high flexibility
Immunomodulatory	6–12	**800–1800**	600–3000	Sequence-dependent; Arg/Lys or Pro/Gly; receptor interaction
Cytoprotectant Anti-ferroptotic	8–18	**1200–2500**	1000–3500	Lipid affinity; redox activity; membrane interaction
Transporter Signaling	8–15	**1200–2200**	800–3000	Arg/Lys Rich; Cell Penetration
Intestinal regenerator	6–12	**800–1600**	600–2000	Pro, Thr, Ser; mucosal interaction

Indicative ranges compiled from published structure–activity studies of food-derived bioactive peptides. eg.: Adapted from Brown & Hancock, 2006; Zou et al., 2016; Santiago-López et al., 2016; Singh et al., 2021; Shen et al., 2025.

Antimicrobial Peptides: Mechanisms of Action

Typical mode of action

1st Electrostatic attraction

Many AMPs are cationic and bind preferentially to negatively charged bacterial surfaces.

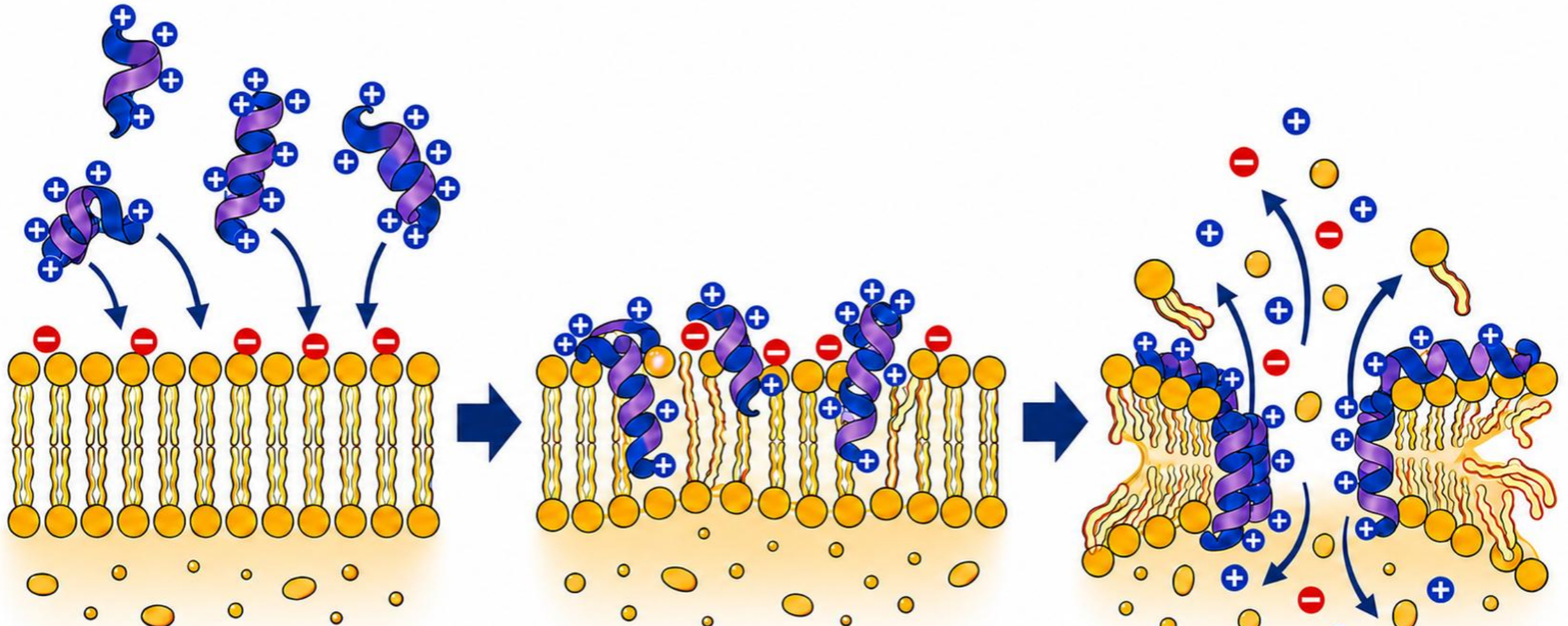
2nd Membrane insertion

Hydrophobic residues penetrate and destabilize the lipid bilayer.

3rd Membrane permeabilization

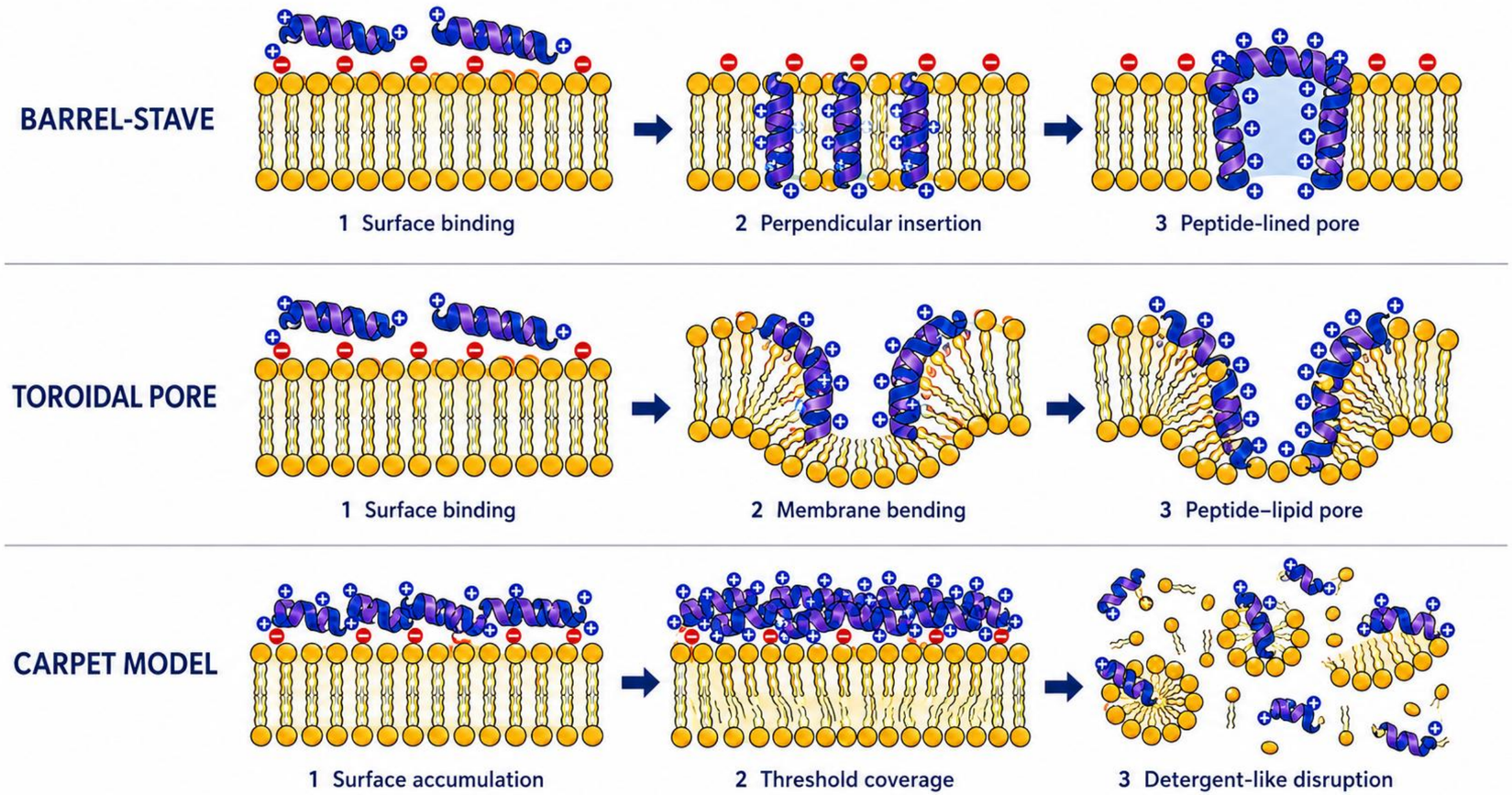
Pore formation or carpet-like disruption causes ion leakage, membrane depolarization and bacterial death.

Some AMPs can also enter the cell and inhibit essential intracellular targets.



Antimicrobial Peptides: Mechanisms of Action

Specific membrane disruption models

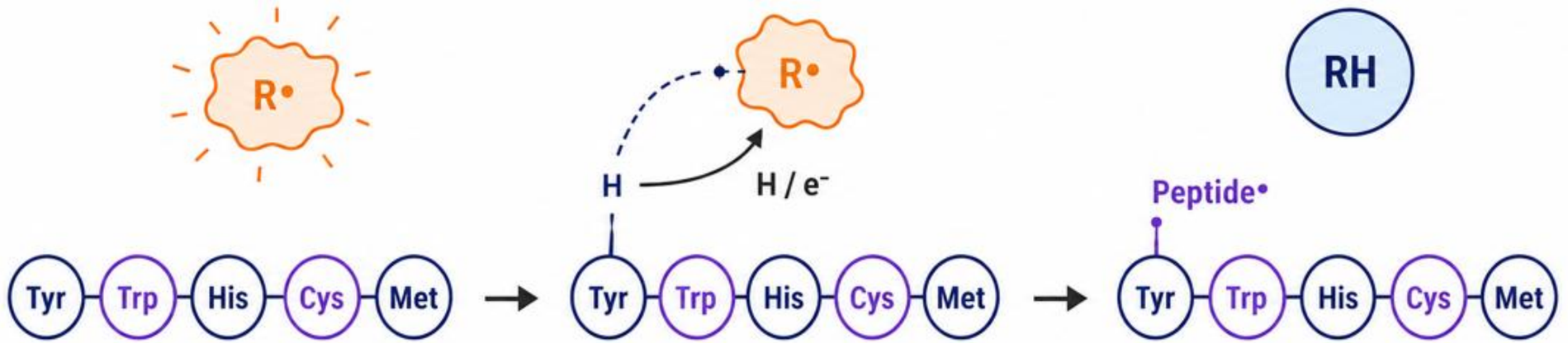


Antioxidant Peptides: Mechanisms of Action

1. Neutralization of Reactive Radicals

Direct scavenging of free radicals.

Some peptides rich in **Tyr, Trp, His, Cys or Met** donate H atoms or electrons to stabilize reactive radicals.



Antioxidant Peptides: Mechanisms of action

2. Limiting Radical Formation and Lipid Peroxidation

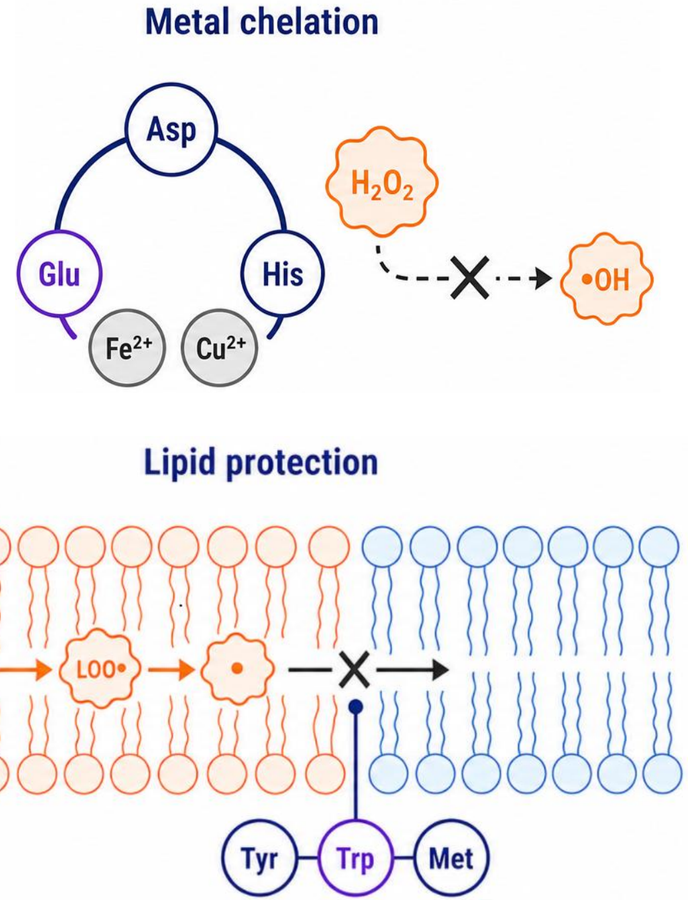
a) Metal Chelation

Some peptides containing **Asp, Glu or His** can bind Fe^{2+} and Cu^{2+} and limit Fenton chemistry.

↓ Fenton reaction → ↓ •OH generation

b) Lipid Protection

Peptides rich in aromatic or hydrophobic residues such as **Tyr, Trp and Met** may **promote peptides interaction with lipid interfaces** and **interrupt lipid-peroxidation chain reactions** → ↓ Membrane damage.



Antioxidant Peptides: Mechanisms of action

3. Activation of Cellular Antioxidant Defenses

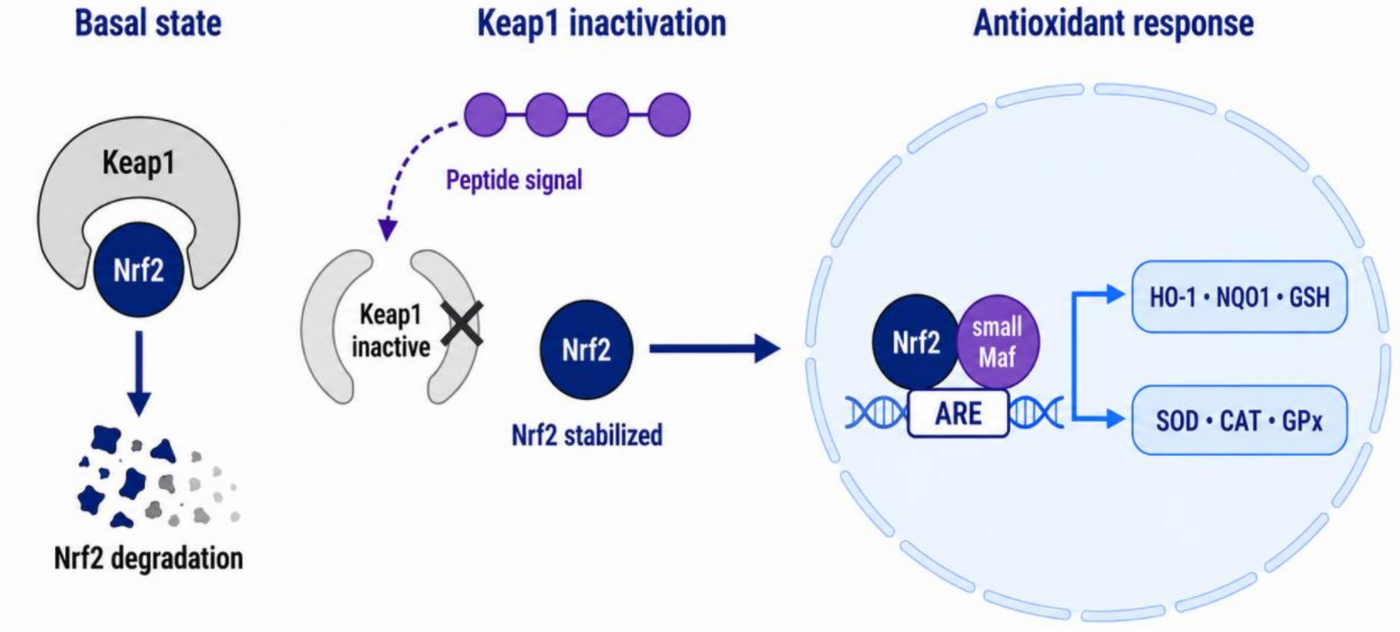
Certain peptides may **inhibit Keap1-mediated degradation of Nrf2**, allowing Nrf2 to accumulate, enter the nucleus, and **activate ARE-dependent antioxidant genes**.(Antioxidant Response Elements).

↑ **HO-1**
(cytoprotective response to stress)

↑ **NQO1**
(detoxification of quinones and oxidizing compounds)

↑ **GSH** synthesis

↑ **SOD, CAT, GPx**
(Endogenous antioxidant capacity)



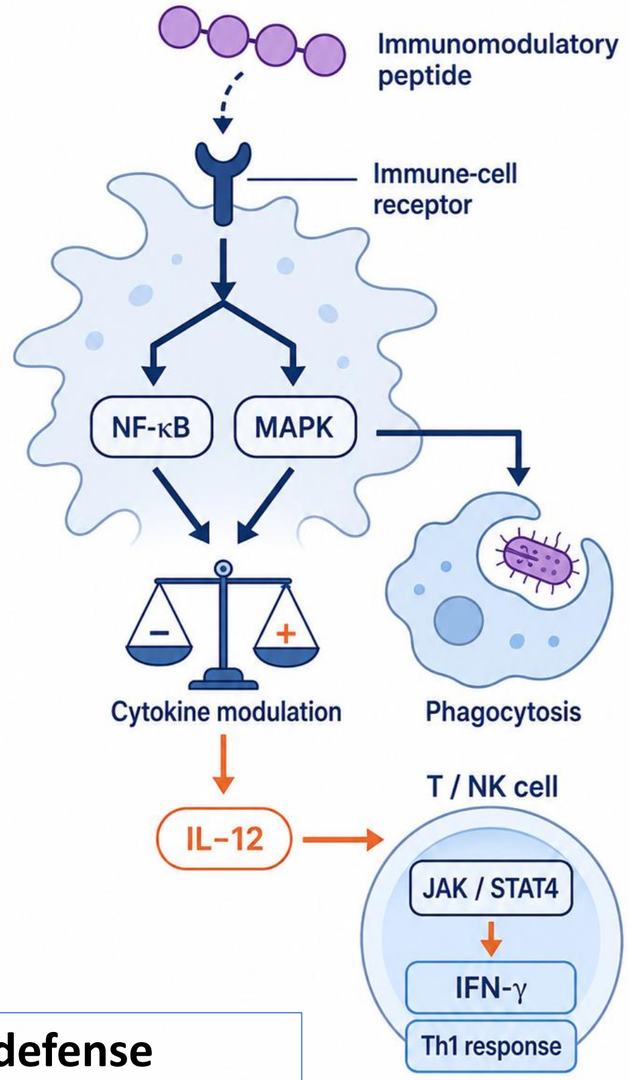
Immunomodulatory Peptides: Mechanisms of action

Depending on their sequence and immune context, some peptides may activate or inhibit pathways such as NF-κB and MAPK.

1st Immune-cell recognition
Some peptide sequences influence receptors on macrophages and other immune cells. e.g., TLR2 and TLR4

2nd Signaling Modulation
NF-κB / MAPK
Sequence- and context-dependent activation or inhibition

3rd Controlled Immune Response
Balanced cytokine production
Enhanced phagocytosis
Coordinated innate and adaptive immunity



The goal, as with β-glucans, is effective defense without excessive inflammation

Example:
Fish Peptides from Enzymatic Hydrolysis



Process (specific example of NBG)



Quality Fish paste

Water

Enzyme

Focused Enzymatic Hydrolysis

- Temperature
- Time
- pH

Enzyme Inactivation

Tricanter

Solids

Fish Oil

>10KDa



Hydrolyzed Fish Paste

Ultrafiltration

Bioactive Peptide

rich paste *

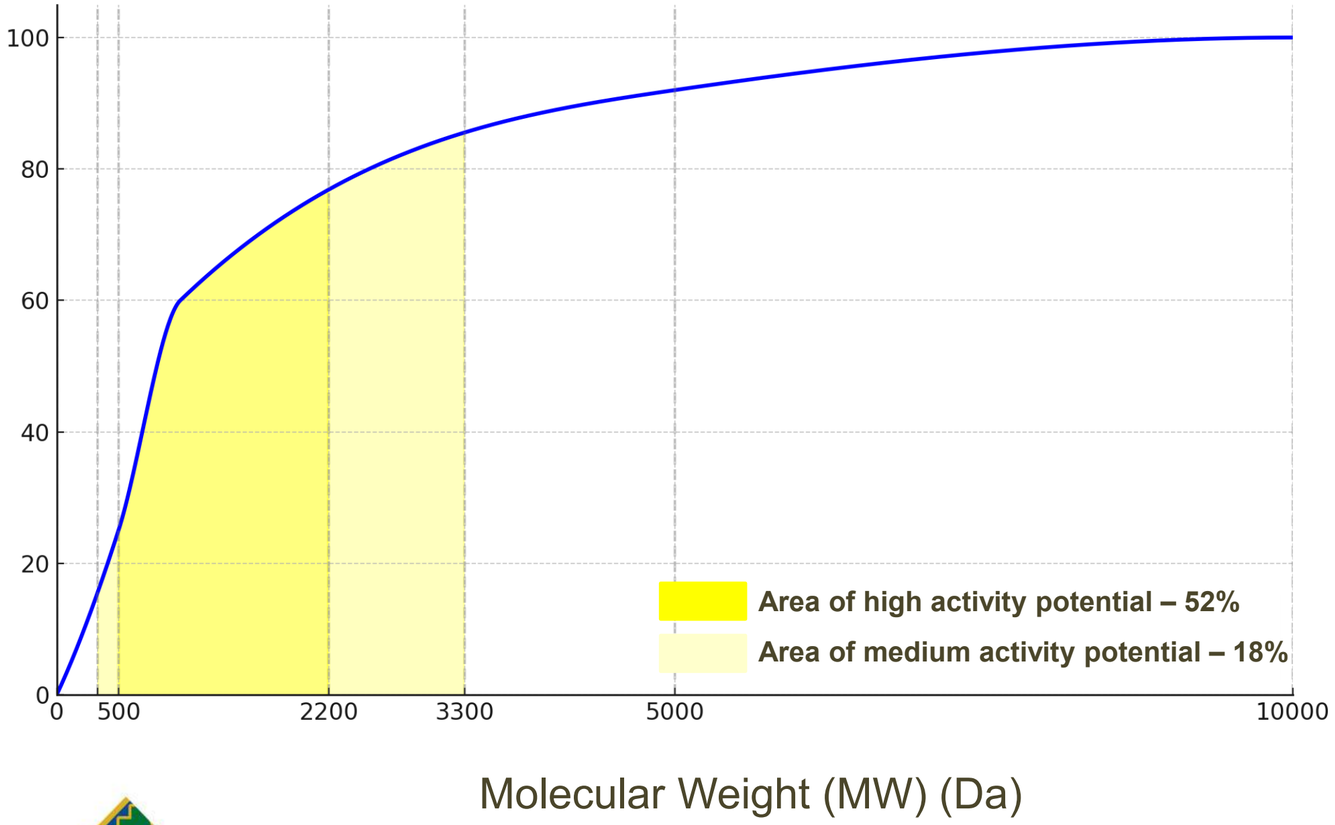
Focused Enzymatic Technology



**** 97.5% of Protein Digestibility***

Molecular weight

Cumulative curve of MW / Fish hydrolysate *(NBG Product)*



✓ **Target** → produce **bioactive peptides**.

✓ **Priority** → **Low and medium molecular weight** peptides.

⊘ Too **much hydrolysis can destroy bioactive peptides**.

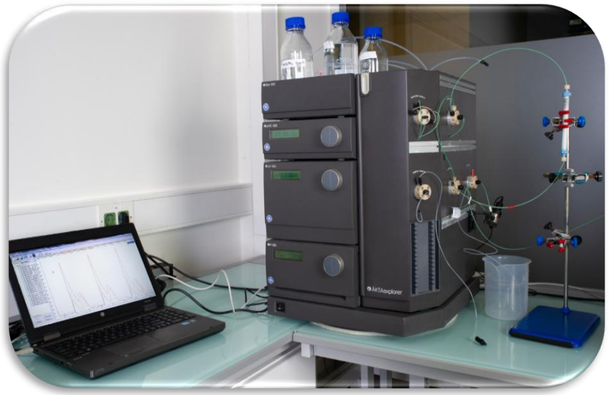
⊘ **Insufficient hydrolysis** produces peptides with **high molecular weight and low bioactivity**.

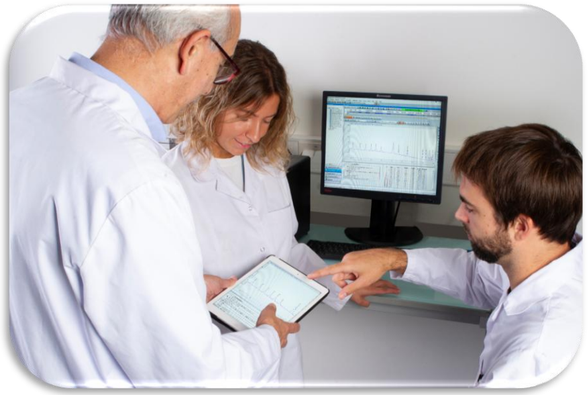


Detection of peptides with Bioactive Potential

Combining a **UHPLC system coupled to an Orbitrap mass Spectrometer** and data list from several sources as Biopep-UWM.

Eg. NBG, from a total of 2800 peptides were identified more than **200 peptides with Bioactive Potential**.

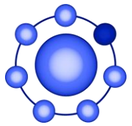




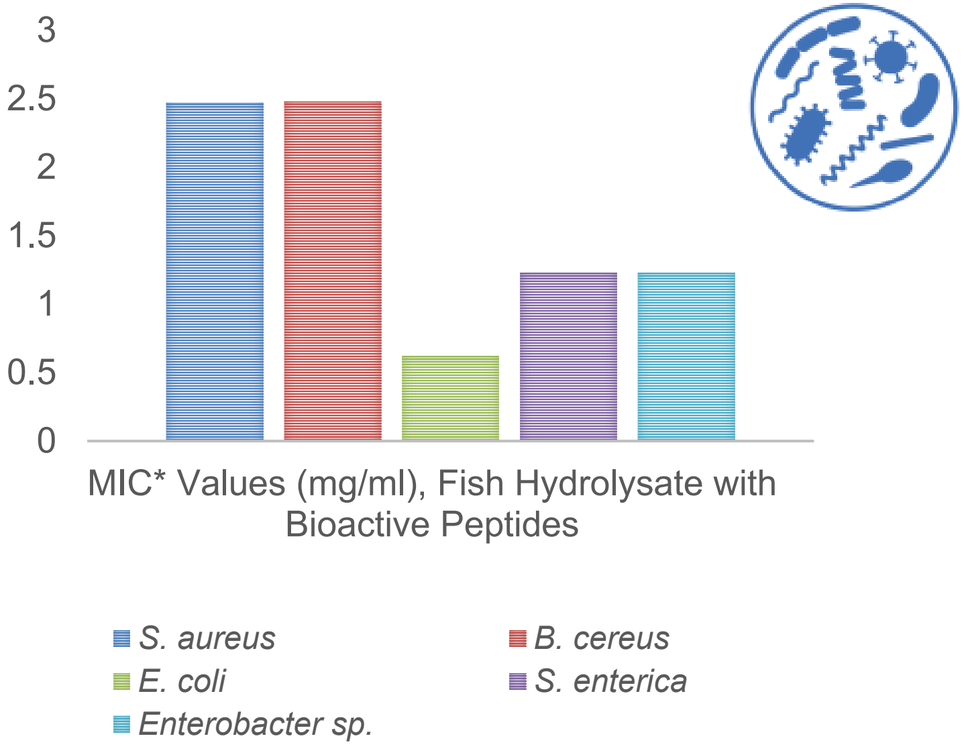
Short example of NBG:

- AGDDAPR
- SKAGAIAGKIA
- IIAPPER
- DNIQGITKPAIR
- IGNLLKSLLX
- IPLMIGAP
- SNKGGGRPN
- FGLLVNPGA
- PFQGVPLGFQAP
- KISGKWKAQK
- LSTIKAFAT
- GWGGLNQTQL

Specific Examples

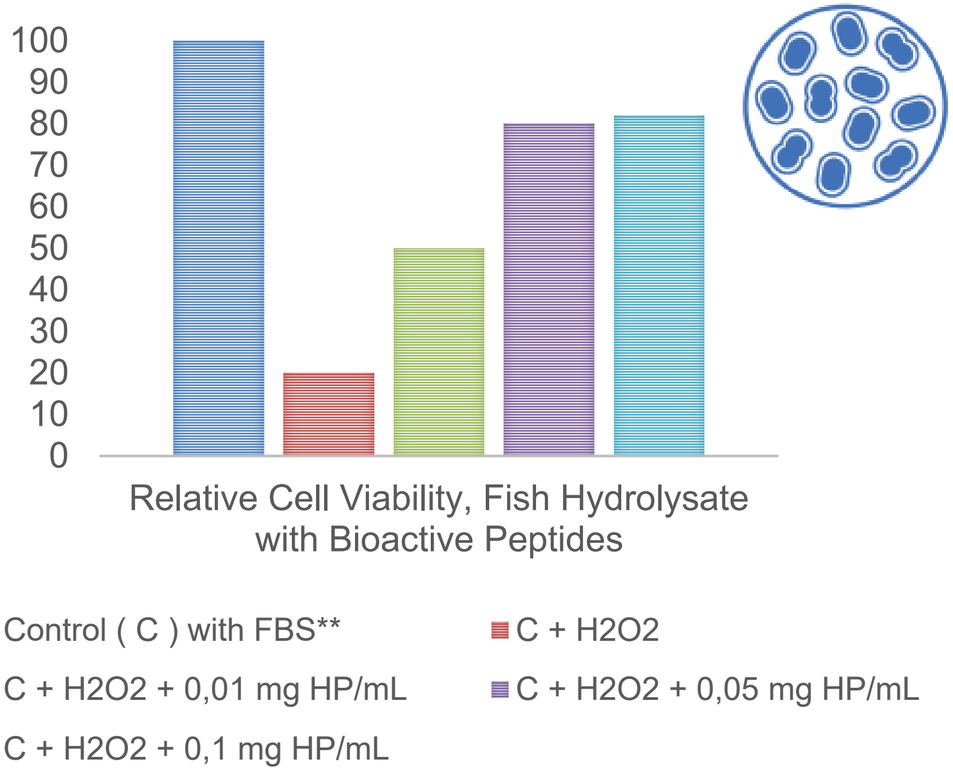
Peptide	Bioactive Potential	Scientific explanation	References
DNIQGITKPAIR	Immunomodulator 	12-aa anti-inflammatory peptide identified in broken-rice protein hydrolysates and also detected in fish hydrolysates. It was selected through peptidomic analysis and in silico MHC-II-binding prediction. In LPS-stimulated macrophages, the synthesized peptide reduced NO production and pro-inflammatory mediators, including TNF-α, IL-6, and IL-1β. A candidate immunomodulatory peptide.	BIOPEP-UWM, *In Vitro Anti-Inflammatory Activity of Three Peptides Derived from the Byproduct of Rice Processing, Qu et al.* (2022)
KISGKWKAQK	Antimicrobial 	10-aa cationic peptide (net charge +4; 40% Lys) consistently predicted as antimicrobial by CAMP_R3. Lys may promote binding to anionic bacterial membranes, while Trp facilitates membrane interaction. Promising in silico antimicrobial candidate.	CAMP_R3 Waghu et al. (2015); Gopal et al., *Amino Acids* (2013)
IIAPPER	Antioxidant 	7-residue multifunctional peptide with experimentally demonstrated in vitro antioxidant activity, including radical scavenging, reducing power, and Fe^{2+} chelation. It also inhibits ACE, α-glucosidase, and pancreatic lipase, enzymes associated with blood-pressure, glucose, and lipid regulation.	BIOPEP-UWM, Zielińska et al., International Journal of Food Science & Technology (2018), Zielińska et al., *Eur Food Res Technol* (2020).

Peptides from fish hydrolysate *in vitro* trials



*MIC: Minimum Inhibitory Concentration

Adapted from Bougatef et al., 2017 (*Food Chemistry*, CSIC–University of Sfax), with permission



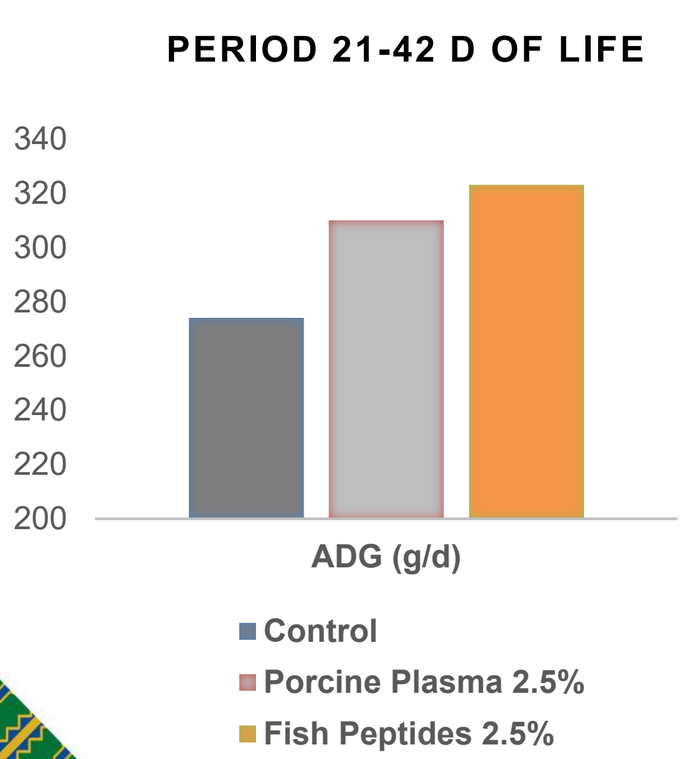
**FBS: Fetal Bovine Serum

Adapted from Giannetto et al., 2020 (*Marine Drugs*, University of Messina, CC BY 4.0)

Peptides from fish hydrolysate *in vivo* trials

Effects Fish peptides Vs. Spray-Dried Plasma on piglet performance

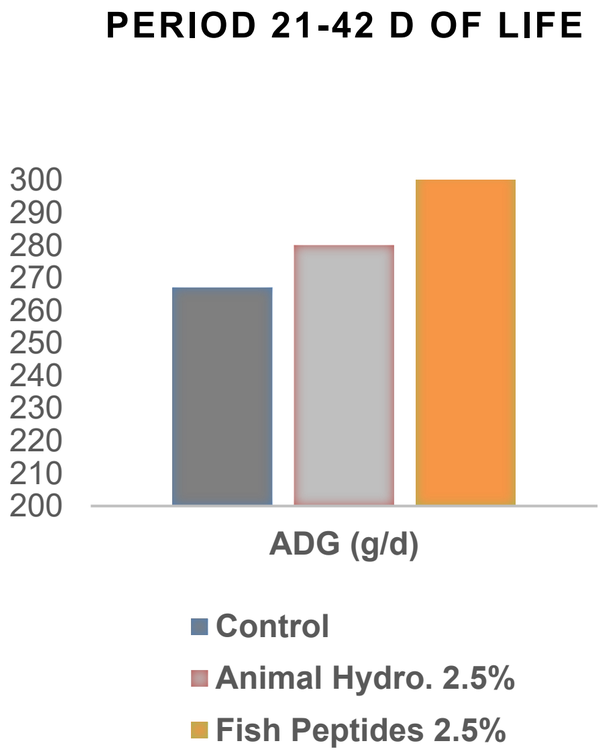
NBG & UBRC, 2022
Philsan, 2024





Effects of Fish Peptides Vs. Regular hydrolyzed fish Protein on piglet performance

NBG & UBRC, 2022
Ciporc, 2026

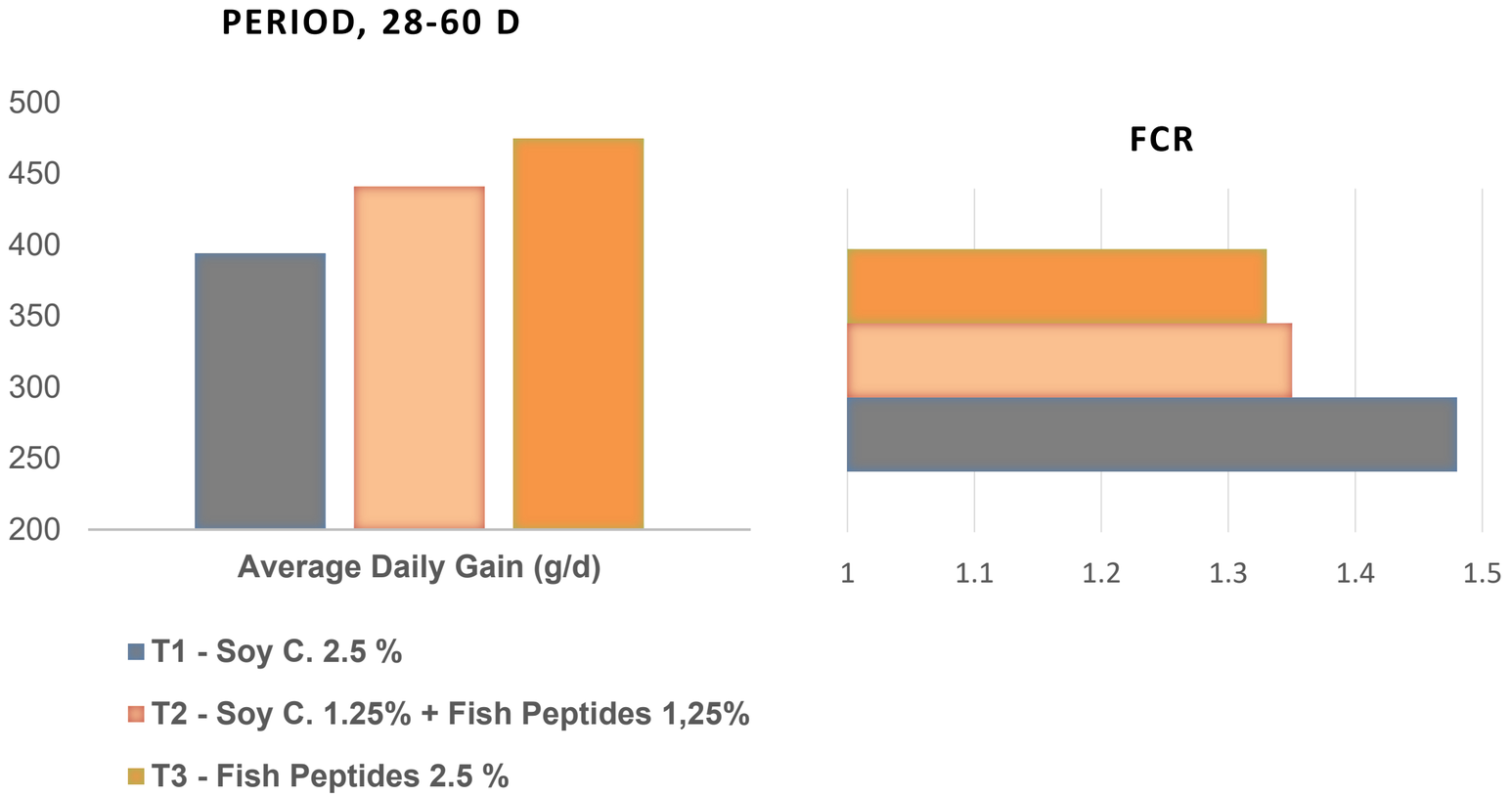


Peptides from fish hydrolysate *in vivo* trials

Effects of Fish Peptides Vs. Soy Protein Concentrate on piglet performance

NBG & Data Omics, 2024
Asia Agri-Tech Expo &Forum 2026



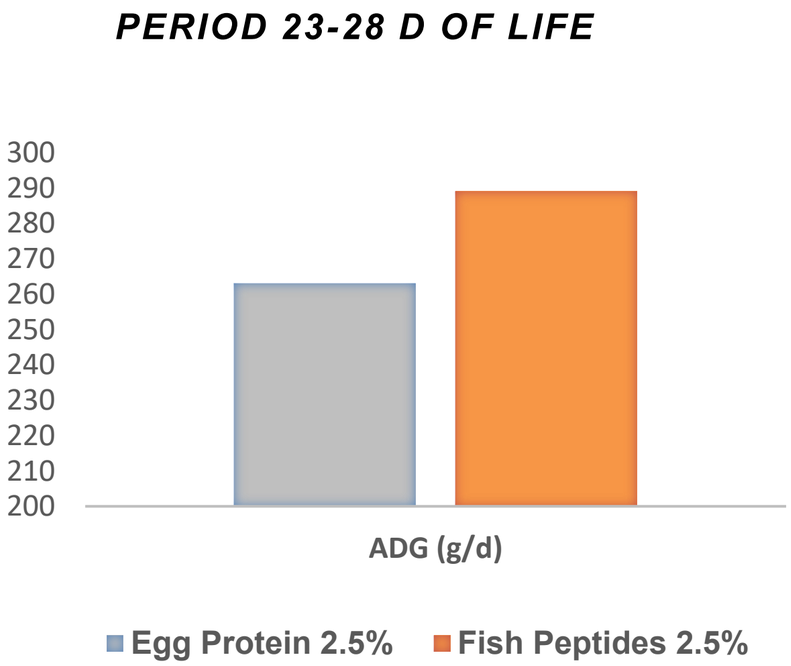


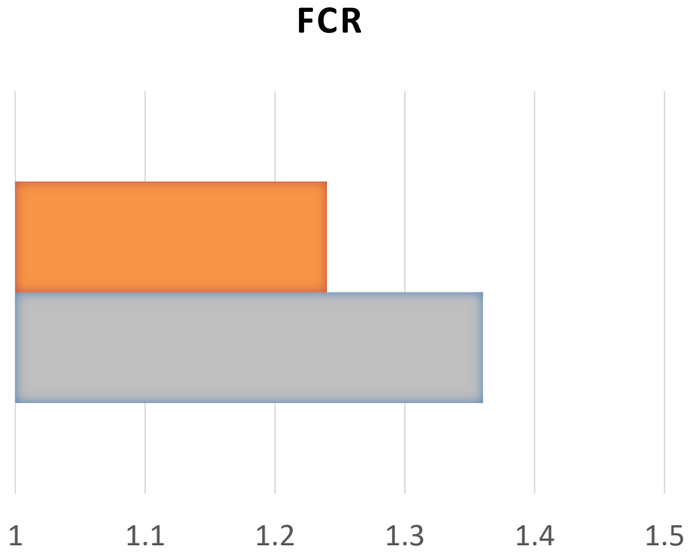
Peptides from fish hydrolysate *in vivo* trials

Effects of Fish Peptides Vs. Egg Meal on piglet performance

NBG, 2023
Ciporc, 2026





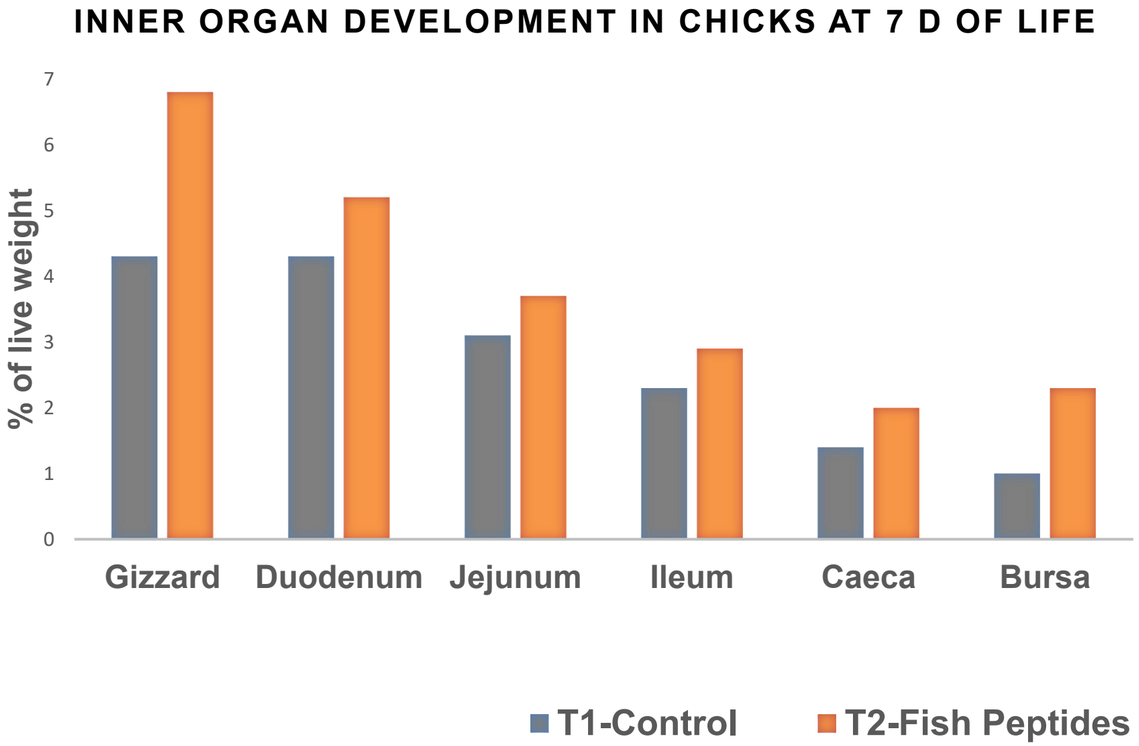


Peptides from fish hydrolysate *in vivo* trials

Effects Fish Peptides in chicks from 0-7 d

NBG & First Ten Consulting, 2024
Philsan, 2024





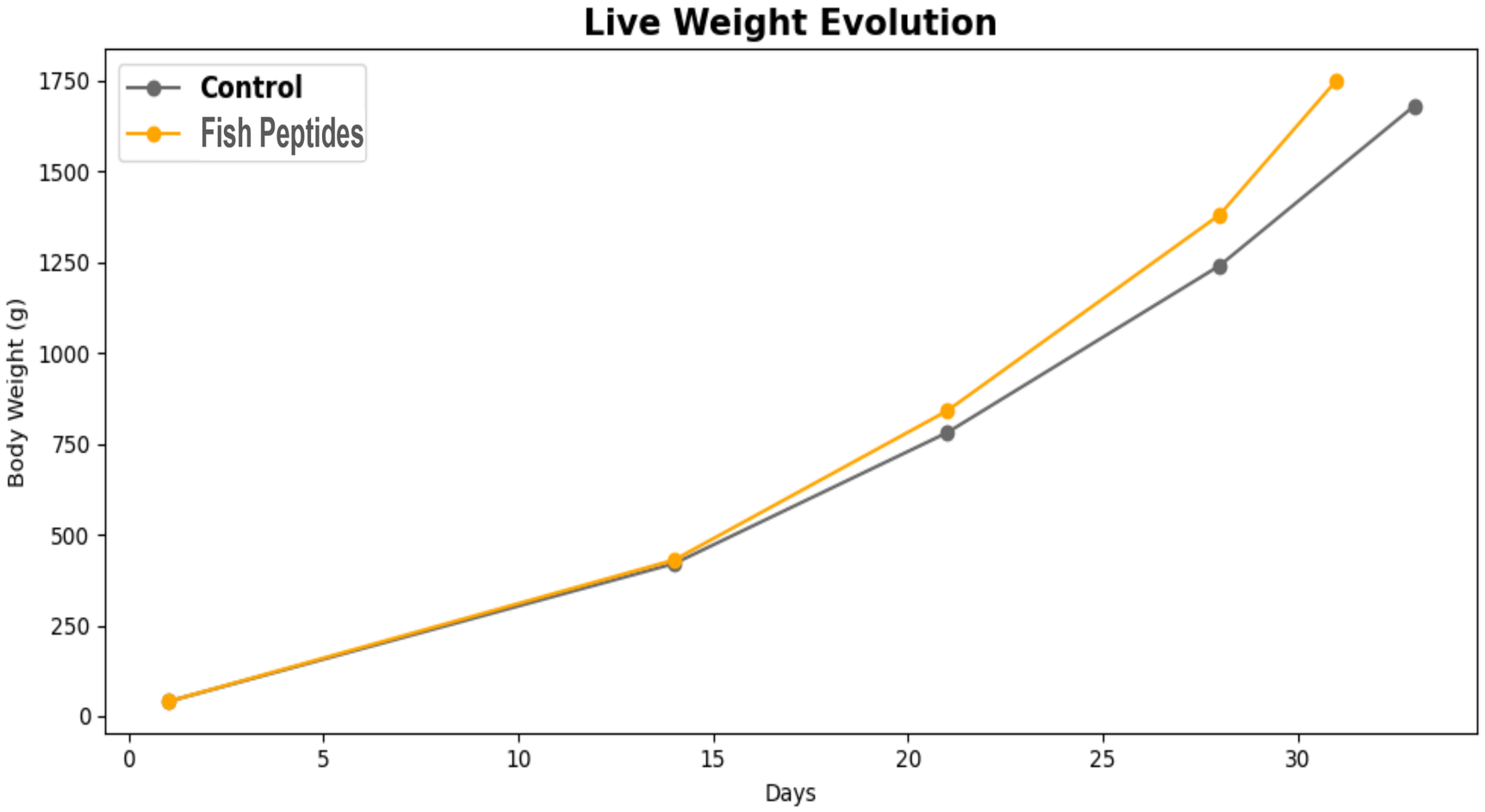
Digestibility at 7 d (%)	T1 / Control	T2 / Fish Peptides
Crude Protein	81.4	**84.7**
Crude Fat	96.6	**97.5**
Crude Energy	85.5	**87**
Calcium	63.8	**68.3**
Phosphorus	85.4	**87.5**

Peptides from fish hydrolysate *in vivo* trials

Effects of Fish Peptides in Broilers

NBG & Data Omics, 2023
Asia Agri-Tech Expo &Forum 2026



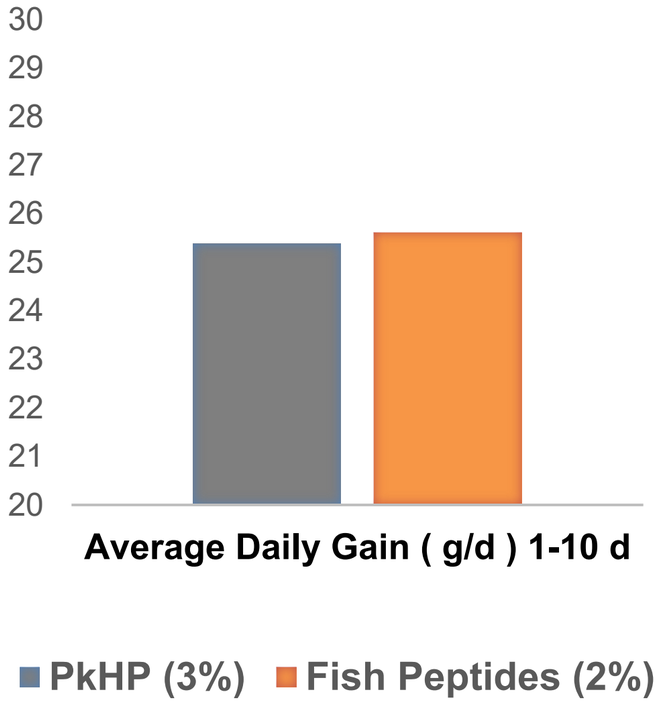


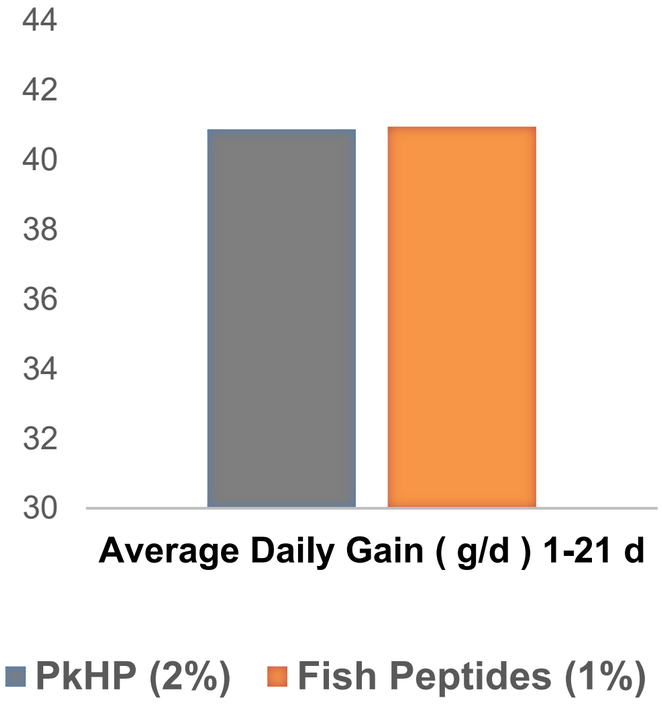
Peptides from fish hydrolysate *in vivo* trials

Effects of Fish Peptides Vs Hydrolyzed Pork protein in Broilers

UBRC & Data Omics, 2025
Asia Agri-Tech Expo &Forum 2026







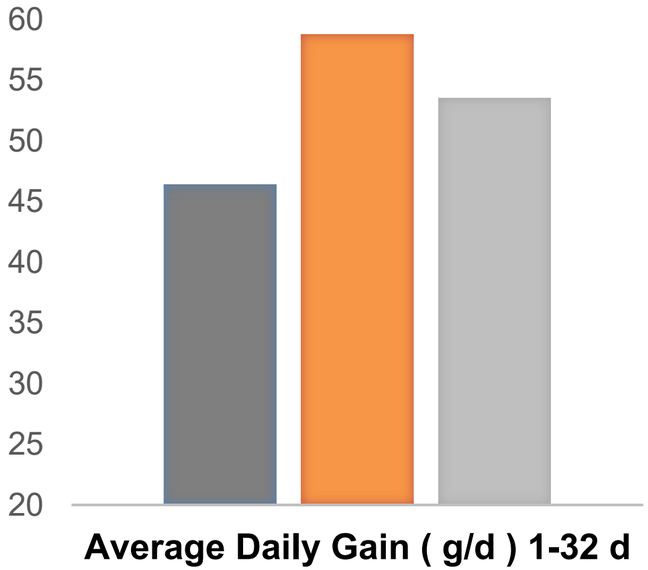
Lower inclusion of optimized fish peptides maintained numerically similar ADG to hydrolyzed pork protein

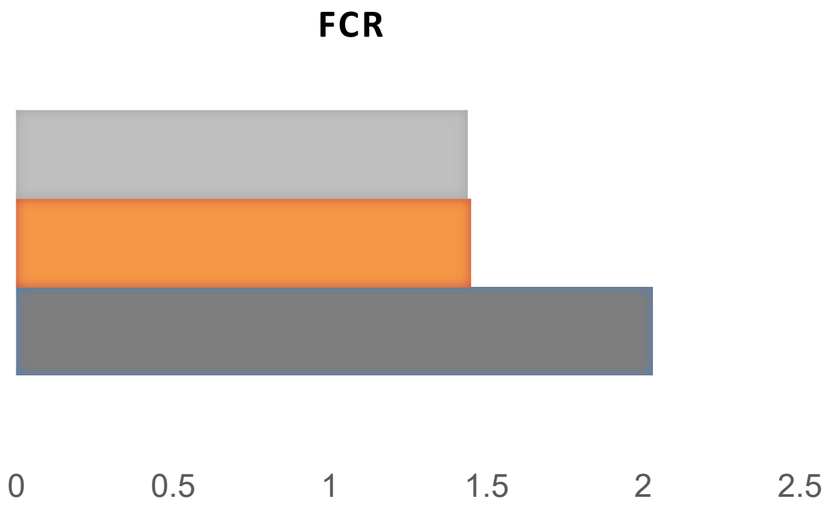
Peptides from fish hydrolysate in vivo trials

Effects of Fish Peptides in broilers when added to a basal diet

UBRC & Data Omics, 2026
New unpublished data





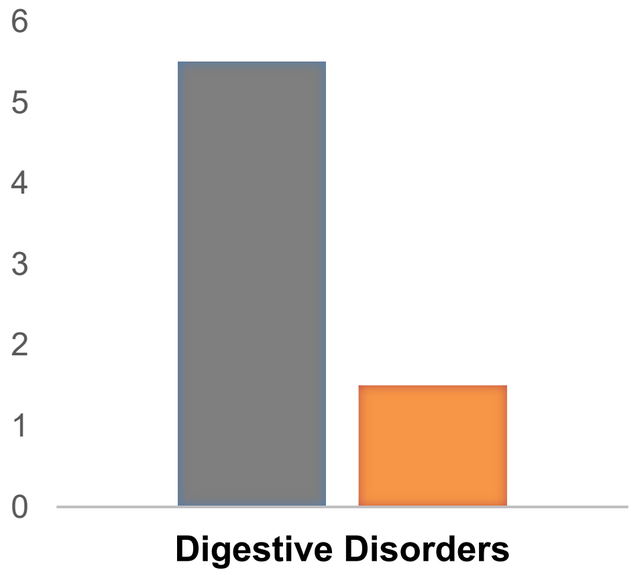


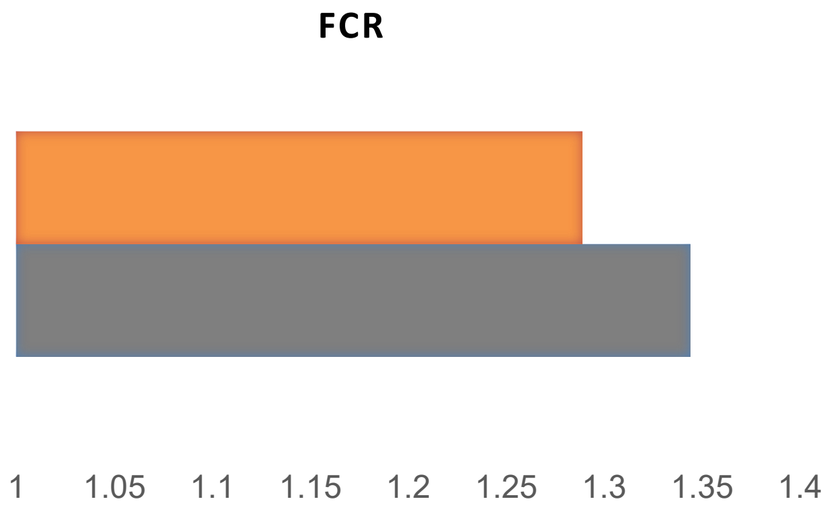
Peptides from fish hydrolysate in vivo trials

Effects of Fish Peptides Vs Poultry Meal in Broilers

RRP Group & NBG, 2026
New unpublished data







Conclusions

- Peptide activity depends on **size, amino acid sequence, charge, and structure**.
- **Source, enzyme selection, and processing** determine the peptide profile.
- **Antimicrobial peptides primarily disrupt microbial membranes**.
- **Antioxidant peptides neutralize ROS** and reinforce cellular defenses.
- **Immunomodulatory peptides regulate immune responses** and inflammation.
- **Peptidomics and databases accelerate candidate discovery** of new Bioactive Peptides.
- **Selected fish peptides show promising functional and performance benefits**.

ROS: Reactive Oxygen Species



NBG digestaPro

NBG enteroPro

NBG _inspired by nature

ADVANCED NATURAL NUTRITION